

nAD1026x2-13d

Dual 10-bit 26 MSPS Analog-to-Digital Converter IP

FEATURES

- TSMC CL013G 1.2 V Technology
- Dual 10-bit Pipeline ADC
- Up to 30 MSPS Conversion Rate
- Excellent Dynamic Performance
 - 58 dBFS SNR at $F_{IN} = 10$ MHz
 - 68 dBc SFDR at $F_{IN} = 10$ MHz
- Dynamic Power Scaling
- Low Power Consumption
 - 16 mW per Channel at 26 MSPS
 - 9.2 mW per Channel at 8 MSPS
- Power Saving Idle Modes
- 1.0 mm² Core Area
- Pure Internal Voltage Reference
- Programmable Input Swing
 - 0.75 to 1.5 V p-p Differential
- Excellent Gain and Phase Matching
- Input Clock Duty Cycle Restorer

APPLICATIONS

- TV / Video / Radio Decoders
 - DAB /DVB
- Quadrature (I and Q) Modulation

GENERAL DESCRIPTION

The nAD1026x2-13d is a monolithic, high-speed, ultra low power, dual channel analog-to-digital converter silicon IP. It uses a fully differential multistage pipeline architecture with digital error correction. The core includes a input PGA and pure internal voltage reference that provides a programmable full-scale range and range of input common voltages. The ADC is designed for high dynamic performance. It thus represents an ideal solution for demanding applications like DVB and DAB.

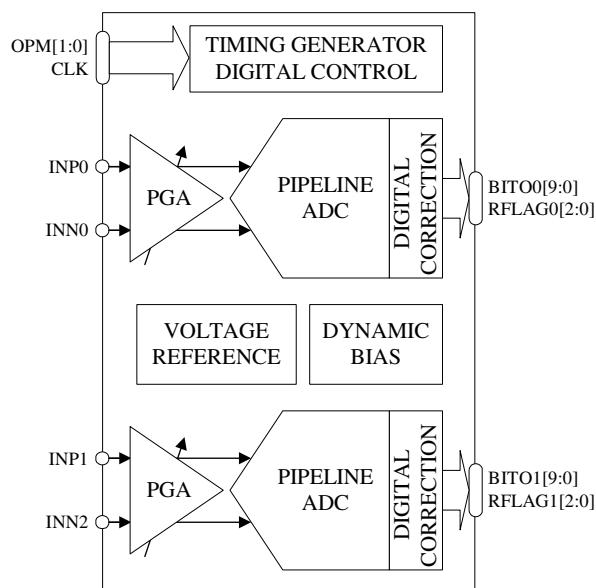


Figure 1. Functional block diagram

The dual ADC consumes only 16 mW at 26 MSPS operation. Dynamic power scaling means that the power consumption is scaled with the operating frequency resulting in only 9.2 mW per channel at 8 MSPS operation. Combined with power saving idle modes the ADC is suitable for battery powered devices. Output data is available in a binary offset coded format. Three out-of-range indicator bits are also available for determining if the input signal is over-range, under-range or out-of-range.

Implemented in a generic 0.13 μ m CMOS process, employing a fully differential architecture and no required external reference decoupling it represents an ideal ADC for highly integrated mixed-signal systems.

QUICK REFERENCE DATA

IP Type / Technology	Hard Macro / TSMC CL013G 1.2 V 1P6M			
IP Area / Dimensions	1.024 mm ² / 1.28 × 0.8 mm			
Parameter	Min.	Typ.	Max.	Unit
Power Dissipation, 26 MSPS		16		mW
Signal-to-Noise Ratio, 26 MSPS $F_{IN} = 10$ MHz		58		dBFS
Spurious-Free-Dynamic Range, 26 MSPS $F_{IN} = 10$ MHz		68		dBc
Effective Number of bits, 26 MSPS $F_{IN} = 10$ MHz		9.3		Bit

Table 1. nAD1026x2-13d quick reference data

**ABOUT NORDIC SEMICONDUCTOR**

Founded in 1983, Nordic Semiconductor ASA (OSE:NOD) is a Fabless Semiconductor specializing in short-range radio communication and high-speed data conversion. Nordic is head quartered in Trondheim, Norway,

MIXED SIGNAL IP OFFERINGS

Leading and emerging Fabless Semiconductor companies around the globe leverage Nordic's portfolio of off-the-shelf mixed signal intellectual property for the design of highly integrated circuits for broadband communication, digital imaging and video.

Our IP portfolio includes medium resolution (8 to 12-bit), high-speed (40 MSPS+) data converters (ADC and DAC) and low jitter clock generation (PLL) implemented in the latest deep submicron CMOS technologies. Featuring low power consumption, low-silicon area, and a minimum of required external components and technology options our products offers cost efficient integration of high-performance data converters in cost sensitive, high volume integrated circuit designs.

DELIVERABLES

Our mixed signal intellectual property is delivered as technology specific hard macros. Upon licensing we provide a complete design-kit and documentation that enables efficient, low risk, design-in and integration. The design kit for our off-the-shelf IPs includes:

- Full datasheet
- Evaluation board and samples
- Silicon validation report
- Integration Guidelines
- Physical design database (flat gds2)
- LVS Netlist (spice compatible)
- Footprint (.lef format)
- HDL model (verilog model)
- Timing model (.lib)
- Design-in and integration support

FURTHER INFORMATION

For further information about this product, such as full datasheet, silicon validation report, availability and licensing terms please contact us at: datacon@nordicsemi.no

DOCUMENT INFO	
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